

## VERSAILLES 3 NNW, KY, USA

WMO: 724300

Lat: 38.0945N

Lon: 84.7465W

Elev: 272

StdP: 98.10

Time Zone: -5.00 (NAE)

Period: 03-19

WBAN: 63838

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |            |            |            | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|------------|------------|------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |            | 1%         |            |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB       | WS         | MCDB       | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-13.3 | (c)<br>-10.4 | (d)<br>-18.5                  | (e)<br>0.8 | (f)<br>-12.0 | (g)<br>-15.9 | (h)<br>1.0 | (i)<br>-10.3 | (j)<br>9.5            | (k)<br>9.3 | (l)<br>8.8 | (m)<br>8.3 | (n)<br>2.3            | (o)<br>N/A | (p)<br>0.405 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.6                    | 32.5            | 23.4 | 31.1 | 23.1 | 30.0 | 22.8 | 25.2                | 30.0 | 24.5 | 29.0 | 23.8 | 28.1 | 3.0                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.7                            | 19.2 | 28.2 | 23.0 | 18.3 | 27.3 | 22.4 | 17.7 | 26.4 | 78.3          | 30.1 | 75.3 | 29.0 | 72.6 | 28.1 | 28.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |      |  |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|------|--|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |      |  |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |      |  |
| (a) 8.0           | (b) 6.9 | (c) 6.1 | (e) -17.2                  | (f) 34.1 | (g) 3.5            | (h) 2.3 | (i) -19.7                                          | (j) 35.7 | (k) -21.8  | (l) 37.1 | (m) -23.8  | (n) 38.4 | (o) -26.3  | (p) 40.1 |      |  |
|                   |         |         | DB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |      |  |
|                   |         |         | WB                         | -17.7    | 26.4               | 4.1     | 0.9                                                | -20.6    | 27.0       | -23.0    | 27.5       | -25.2    | 28.0       | -28.2    | 28.6 |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 13.1        | 0.6   | 2.3   | 7.5   | 13.3  | 18.5  | 22.5  | 23.9  | 23.5  | 20.3  | 13.9  | 7.6   | 3.2   |  |  |
|                                                                        | DBStd       | 9.66        | 6.82  | 6.50  | 6.26  | 5.04  | 4.41  | 2.84  | 2.50  | 2.70  | 3.79  | 5.02  | 5.37  | 5.62  |  |  |
|                                                                        | HDD10.0     | 1029        | 301   | 227   | 126   | 26    | 2     | 0     | 0     | 0     | 0     | 18    | 107   | 222   |  |  |
|                                                                        | HDD18.3     | 2532        | 551   | 450   | 339   | 162   | 56    | 4     | 1     | 2     | 23    | 154   | 323   | 469   |  |  |
|                                                                        | CDD10.0     | 2174        | 8     | 11    | 47    | 124   | 264   | 376   | 430   | 418   | 309   | 140   | 36    | 11    |  |  |
|                                                                        | CDD18.3     | 636         | 0     | 0     | 2     | 9     | 60    | 129   | 172   | 162   | 82    | 17    | 1     | 0     |  |  |
|                                                                        | CDH23.3     | 5036        | 0     | 1     | 10    | 62    | 414   | 1024  | 1396  | 1308  | 684   | 133   | 3     | 0     |  |  |
|                                                                        | CDH26.7     | 1549        | 0     | 0     | 0     | 2     | 76    | 307   | 468   | 445   | 223   | 28    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.7         | 3.5   | 3.4   | 3.4   | 3.1   | 2.2   | 2.2   | 2.0   | 1.8   | 2.0   | 2.4   | 2.7   | 3.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1232        | 88    | 81    | 115   | 115   | 133   | 115   | 127   | 95    | 89    | 83    | 83    | 108   |  |  |
|                                                                        | PrecMax     | 1664        | 135   | 143   | 355   | 342   | 269   | 219   | 240   | 217   | 233   | 166   | 197   | 254   |  |  |
|                                                                        | PrecMin     | 827         | 30    | 33    | 41    | 40    | 26    | 28    | 45    | 33    | 19    | 15    | 25    | 44    |  |  |
|                                                                        | PrecStd     | 188         | 29    | 33    | 61    | 64    | 62    | 46    | 55    | 42    | 54    | 47    | 40    | 46    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 18.3  | 20.5  | 24.7  | 26.9  | 30.3  | 32.8  | 34.4  | 34.5  | 33.7  | 29.3  | 23.3  | 19.0  |  |  |
|                                                                        |             | MCWB        | 14.6  | 15.4  | 17.8  | 17.8  | 22.4  | 23.0  | 23.2  | 24.7  | 21.8  | 20.0  | 16.7  | 15.4  |  |  |
|                                                                        | 2%          | DB          | 16.1  | 17.5  | 21.9  | 25.3  | 28.6  | 31.0  | 32.0  | 32.4  | 31.1  | 26.7  | 20.6  | 16.5  |  |  |
|                                                                        |             | MCWB        | 12.7  | 13.5  | 15.3  | 17.6  | 21.2  | 23.0  | 24.3  | 23.4  | 21.7  | 19.4  | 14.5  | 13.7  |  |  |
|                                                                        | 5%          | DB          | 14.2  | 15.1  | 19.4  | 23.5  | 27.4  | 29.9  | 30.7  | 30.7  | 29.3  | 24.5  | 18.1  | 14.3  |  |  |
|                                                                        |             | MCWB        | 12.0  | 12.0  | 13.9  | 16.5  | 20.4  | 22.5  | 23.9  | 23.1  | 21.3  | 17.8  | 13.8  | 12.4  |  |  |
|                                                                        | 10%         | DB          | 11.5  | 12.6  | 17.1  | 21.6  | 25.9  | 28.6  | 29.5  | 29.3  | 27.4  | 22.0  | 16.1  | 12.2  |  |  |
|                                                                        |             | MCWB        | 8.9   | 9.6   | 12.4  | 15.5  | 20.1  | 22.0  | 23.4  | 22.6  | 20.7  | 17.2  | 12.0  | 10.1  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 15.3  | 16.7  | 18.2  | 20.1  | 23.4  | 25.2  | 26.7  | 25.8  | 24.3  | 22.9  | 17.6  | 16.6  |  |  |
|                                                                        |             | MCDB        | 17.8  | 20.4  | 23.8  | 25.2  | 28.4  | 30.4  | 31.0  | 31.2  | 29.1  | 27.8  | 21.0  | 18.8  |  |  |
|                                                                        | 2%          | WB          | 13.2  | 15.2  | 16.5  | 18.8  | 22.3  | 24.4  | 25.5  | 24.7  | 23.4  | 21.2  | 15.9  | 14.4  |  |  |
|                                                                        |             | MCDB        | 15.0  | 17.5  | 20.7  | 23.7  | 27.1  | 29.2  | 30.3  | 29.2  | 28.7  | 24.6  | 18.7  | 16.1  |  |  |
|                                                                        | 5%          | WB          | 10.7  | 13.3  | 14.9  | 17.6  | 21.6  | 23.6  | 24.8  | 23.9  | 22.5  | 19.6  | 14.0  | 12.8  |  |  |
|                                                                        |             | MCDB        | 12.6  | 15.3  | 18.2  | 21.6  | 26.3  | 28.3  | 29.4  | 28.2  | 27.3  | 22.7  | 16.6  | 14.5  |  |  |
|                                                                        | 10%         | WB          | 8.1   | 10.7  | 13.2  | 16.4  | 20.8  | 22.8  | 24.1  | 23.2  | 21.7  | 17.5  | 12.0  | 10.8  |  |  |
|                                                                        |             | MCDB        | 10.1  | 13.0  | 16.2  | 20.5  | 25.1  | 27.3  | 28.2  | 27.2  | 25.8  | 21.5  | 15.0  | 12.7  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.5   | 9.1   | 10.0  | 10.7  | 9.9   | 10.1  | 9.6   | 10.1  | 10.8  | 10.6  | 9.8   | 8.0   |  |  |
|                                                                        |             | MCDBR       | 10.2  | 12.1  | 11.9  | 12.4  | 11.1  | 11.5  | 11.2  | 11.8  | 13.0  | 12.7  | 11.5  | 10.3  |  |  |
|                                                                        | 5% WB       | MCWBR       | 9.3   | 9.4   | 7.8   | 7.7   | 5.5   | 5.2   | 4.6   | 4.7   | 5.1   | 6.0   | 7.6   | 8.3   |  |  |
|                                                                        |             | MCWBR       | 10.2  | 11.5  | 10.3  | 10.9  | 9.8   | 10.0  | 9.4   | 9.2   | 10.2  | 10.0  | 9.6   | 9.0   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.309 | 0.322 | 0.352 | 0.392 | 0.437 | 0.449 | 0.473 | 0.467 | 0.412 | 0.355 | 0.322 | 0.300 |  |  |
|                                                                        |             | taud        | 2.464 | 2.406 | 2.386 | 2.271 | 2.190 | 2.194 | 2.136 | 2.145 | 2.271 | 2.429 | 2.476 | 2.491 |  |  |
|                                                                        | Ebn at Noon |             | 874   | 906   | 909   | 886   | 849   | 835   | 811   | 806   | 833   | 857   | 850   | 859   |  |  |
|                                                                        |             | Edh at Noon | 81    | 98    | 111   | 132   | 145   | 145   | 153   | 148   | 123   | 95    | 80    | 74    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.92  | 2.61  | 3.70  | 4.93  | 5.52  | 6.18  | 5.84  | 5.49  | 4.53  | 3.38  | 2.33  | 1.66  |  |  |
|                                                                        | RadStd      |             | 0.21  | 0.37  | 0.34  | 0.36  | 0.51  | 0.33  | 0.33  | 0.31  | 0.47  | 0.38  | 0.26  | 0.17  |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (25 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +122    | +76     |

Nomenclature: See separate page